ESP32 + SD Module — Round 2 Powered Electrical Tests

⚠ WARNING: USB POWER MUST BE CONNECTED FOR ALL ROUND 2 TESTS. USE DC VOLTAGE MODE ONLY

ESP32 and SD module are POWERED. NEVER use resistance (Ω) or current (A / mA) mode on the powered circuit.

Multimeter setup: Mode = DC Voltage (DCV) | Black probe → COM | Red probe → VΩmA | Keep probe tips off adjacent pins

Date	03-10-2026	Multimeter	DT-830B
ESP32 board		ESP32 activity during P3–P6	
SD module		Firmware / sketch	

Fill in: Actual Reading, Pass/Fail (P1–P2 only), Notes, and the summary. Grey Pass/Fail cells (P3–P6) are record-only: do not mark PASS/FAIL.

Test #	Test Description	Red Probe	Black Probe	USB Connected?	Meter Mode	Expected Reading	Actual Reading	Pass/Fail	Notes
P1	SD module supply voltage	SD module 3.3V	SD module GND	YES	DCV	Approximately 3.3 V	3.27V	✓	Investigate (port wrong)
P2	ESP32 3V3 supply voltage	ESP32 3V3	ESP32 GND	YES	DCV	Approximately 3.3 V	3.27V	✓	
P3	MISO voltage relative to GND	SD module MISO	SD module GND	YES	DCV	Record actual voltage; this is signal-dependent. Do not apply a fixed pass/fail threshold.	1.025V		Fluctuates
P4	MOSI voltage relative to GND	SD module MOSI	SD module GND	YES	DCV	Record actual voltage; this is signal-dependent.	0.1 ∠		Not much really
P5	SCK voltage relative to GND	SD module SCK	SD module GND	YES	DCV	Record actual voltage; this is signal-dependent.	0.1 ✓		Low Idle
P6	CS voltage relative to GND	SD module CS	SD module GND	YES	DCV	Record actual voltage; this is signal-dependent.	3.26V		High / Idle normal

SUMMARY

Measurement	Value (from table)	Comment
SD supply voltage		
ESP32 3V3 voltage		
MISO measured voltage		
MOSI measured voltage		
SCK measured voltage		
CS measured voltage		
General observations	Other than the fluctuations, most of the measurements/test fall within normal parameters.	

UNUSUAL OBSERVATIONS DURING TESTING (heat, odd smells, resets, flickering readings, loose connections, etc.)

None

Summary values link to the Actual Reading column; enter readings in volts (e.g. 3.28) or overwrite the summary cells by hand.